# Project Design Phase-II

## Technology Stack (Architecture & Stack)

**Date:** 02 October 2026
**Team ID:** SWTID-2026-9980
**Project Name:** Script-Controlled ACL – Restrict Record Access Based on Field Value
**Maximum Marks:** 4 Marks

## Technical Architecture:

The project is implemented on the ServiceNow platform to restrict access to records based on a field value. Users access the Institution Details table through the ServiceNow UI. When a user performs Read, Create, Write, or Delete operations, the corresponding ACL checks the user's role and executes a server-side script. The script evaluates the selected record field, such as Branch, and permits or denies access according to the configured condition.

### Architecture Flow:

**User → ServiceNow UI → Record ACL → Role Validation → Field-Value Script → Allow / Deny → Institution Details Record**

**Architecture components:**

- User Interface: ServiceNow forms and lists.
- Application Logic: Record-level ACLs and server-side JavaScript.
- Security: Custom roles combined with field-value based conditions.
- Data Storage: Institution Details records in the ServiceNow platform database.
- External Interfaces: No external third-party API is required for the core functionality.
- Machine Learning: Not applicable; access decisions are rule-based.
- Infrastructure: ServiceNow cloud platform / ServiceNow development instance.

## Table-1 : Components & Technologies:

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | User Interface | ServiceNow forms and lists used to view and manage Institution Details records. | ServiceNow UI |
| 2 | Application Logic-1 | Checks whether the logged-in user has the required custom role before allowing the operation. | ServiceNow ACL + Roles |
| 3 | Application Logic-2 | Checks a record field value through a script and restricts access when the condition is not satisfied. | Server-side JavaScript / ACL Script |
| 4 | Application Logic-3 | Separate ACLs control Read, Create, Write and Delete operations. | ServiceNow Record ACLs |

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|---|---|---|---|
| 5 | Database | Stores Institution Details records and fields such as Branch. | ServiceNow Platform Database |
| 6 | Cloud Database | Managed data storage provided by the ServiceNow platform. | ServiceNow Cloud Data Store |
| 7 | File Storage | No dedicated file storage is required for the core ACL functionality. | ServiceNow Attachment Storage (if required) |
| 8 | External API-1 | No external API is required for field-value based access control. | Not Applicable |
| 9 | External API-2 | No external API is required for the ACL decision. | Not Applicable |
| 10 | Machine Learning Model | Not used. Decisions are based on roles, ACL rules and field values. | Not Applicable |
| 11 | Infrastructure (Server / Cloud) | ACL rules and application logic run within the ServiceNow instance. | ServiceNow Cloud Platform |

## Table-2: Application Characteristics:

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1 | Open-Source Frameworks | No separate open-source framework is required. The project uses native ServiceNow ACL and scripting capabilities. | ServiceNow Platform / JavaScript |
| 2 | Security Implementations | Restricts record operations using ACLs, custom roles and a script-based field-value condition. | ServiceNow ACL, Roles, Server-side JavaScript |
| 3 | Scalable Architecture | The ACL pattern can be applied to additional tables, fields and operations as the application grows. | ServiceNow ACL Framework |
| 4 | Availability | The solution operates within the managed ServiceNow platform and is accessed through the ServiceNow instance. | ServiceNow Cloud Platform |
| 5 | Performance | ACL checks are kept focused on the required role and field-value condition to avoid unnecessary processing. | ServiceNow ACL / JavaScript |

## References:


• ServiceNow Documentation – Access Control Lists (ACLs)

• ServiceNow Documentation – Roles and User Access

• ServiceNow Developer Documentation – Server-side JavaScript

• ServiceNow Developer Documentation – Security and Access Controls